Mutations in the Long-Term Evolution Experiment

Variant calling on three *E. coli* clones from population Ara-3

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Question

How many mutations did *E. coli* accumulate over 50,000 generations of the Long-Term Evolution Experiment [1], and is that answer robust to how strictly we filter the variant calls?

Data

```
# Number of reads per file used for the length, GC and quality summaries.
# Change it without editing the report: quarto render ... -P max_reads:5000
max_reads = 10000
```

```
import sys

import matplotlib
import matplotlib.pyplot as plt
import pandas as pd

from scripts.fastq_stats import quality_by_position, summarize

samples = pd.read_csv("metadata/samples.tsv", sep="\t")
fastq_files = list(samples["fastq_1"]) + list(samples["fastq_2"])
```

We use a subset of the reads prepared by Data Carpentry [2]: three clones of population Ara-3, frozen at generations 5,000, 15,000, 50,000 (Table 1).

```
samples[["sample", "strain", "generation", "mutator", "cit"]]
```

Table 1: The three LTEE clones in this project.

	sample	strain	generation	mutator	cit
0	SRR2589044	REL2181A	5000	NaN	unknown
1	SRR2584863	REL7179B	15000	NaN	unknown
2	SRR2584866	REL11365	50000	plus	plus

Reads

```
reads = pd.DataFrame([summarize(f, max_reads) for f in sorted(fastq_files)])
reads.round(1)
```

Table 2: Reads per FASTQ file. Length, GC content and mean base quality are computed on the first reads of each file (parameter `max_reads`).

	file	reads	mean_length	gc_percent	mean_quality
0	SRR2584863_1.trim.sub.fastq	175000	130.4	50.5	35.4
1	SRR2584863_2.trim.sub.fastq	175000	121.7	50.5	34.8
2	SRR2584866_1.trim.sub.fastq	175000	128.5	50.5	35.2
3	SRR2584866_2.trim.sub.fastq	175000	130.6	50.7	35.5
4	SRR2589044_1.trim.sub.fastq	175000	130.1	50.5	35.5
5	SRR2589044_2.trim.sub.fastq	175000	123.1	50.4	34.9

Each file has 175,000 reads, so each sample has 175,000 read pairs. The mean read length is 127 bases because the reads were trimmed before subsampling. Figure 1 shows that base quality stays high along the reads.

```
colors = ["#2a78d6", "#eb6834", "#1baf7a"]  # one per sample, in generation order
fig, ax = plt.subplots(figsize=(7, 4.2))
for color, (_, row) in zip(colors, samples.sort_values("generation").iterrows()):
    for read, style in [("fastq_1", "-"), ("fastq_2", "--")]:
        quality_by_position(row[read], max_reads).plot(
            ax=ax, color=color, linestyle=style, linewidth=1.5,
            label=f"{row['sample']} (gen. {row['generation']:,}) read {read[-1]}")
ax.set_xlabel("Position in read (bases)")
ax.set_ylabel("Mean Phred quality")
ax.grid(alpha=0.3)
ax.legend(fontsize=7, ncol=3, frameon=False, loc="upper center",
bbox_to_anchor=(0.5, -0.18))
```

```
fig.tight_layout()
plt.show()
```

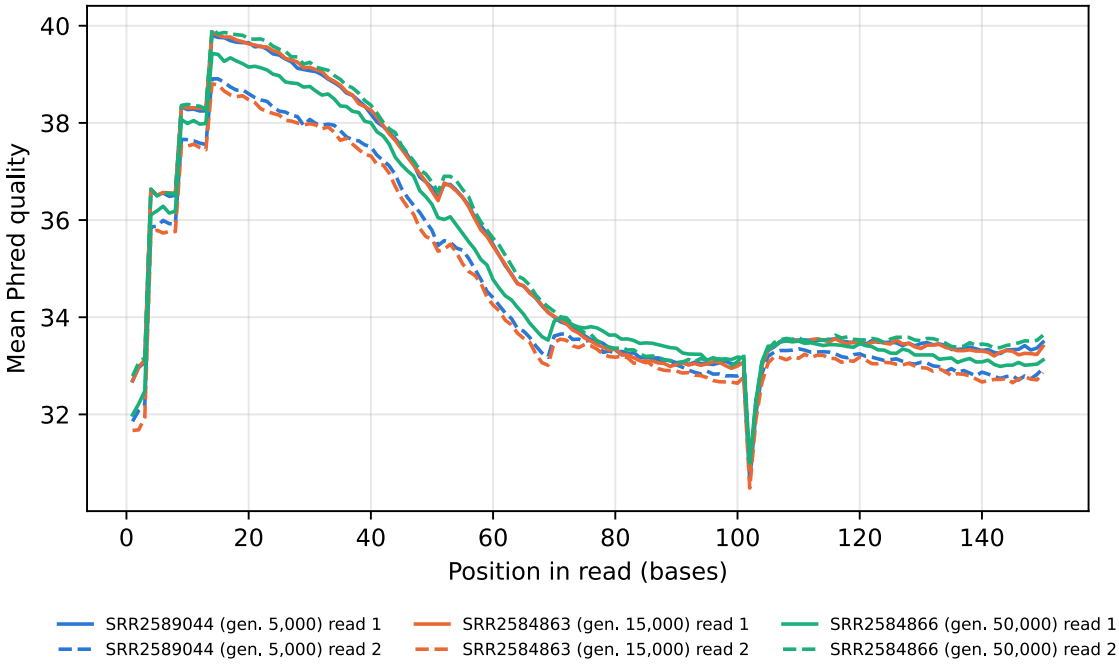


Figure 1: Mean base quality along the reads (first `max_reads` reads of each file). Colour shows the sample, line style the read of the pair.

Software

```
print(f"Python {sys.version.split()[0]}, pandas {pd.__version__}, matplotlib
{matplotlib.__version__}")
```

```
Python 3.13.18, pandas 3.1.0, matplotlib 3.11.2
```

The exact versions of every package are in `pixi.lock`.

References

Bibliography

[1] O. Tenaillon *et al.*, "Tempo and mode of genome evolution in a 50,000-generation experiment," *Nature*, vol. 536, pp. 165–170, 2016, doi: 10.1038/nature18959.

[2] O. Tenaillon, J. E. Barrick, N. Ribeck, and others, "Data Carpentry Genomics Curriculum Example Data." figshare, 2019. doi: 10.6084/m9.figshare.7726454.v3.